

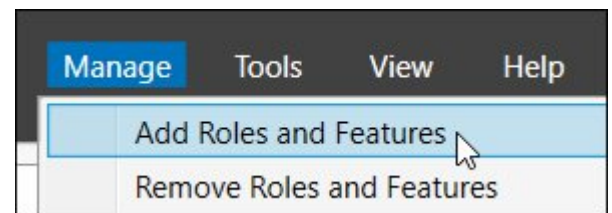
Overview

This lab covers the steps to configure a simple Active Directory setup. There is some assumed prerequisite knowledge and so default/common steps may be omitted for brevity. That said the process should be straightforward and relatively easy to follow.

ADDS Server Setup

Operating System: Windows server 2022

Before we can start using Active Directory it needs to be installed on the server. Using Server Manager Dashboard we can navigate to Manage>Add Roles and Features. Clicking this will start the Add Roles and Features wizard. On the following screen we'll choose Role-based or feature-based installation.



☒ **Role-based or feature-based installation**

Configure a single server by adding roles, role services, and features.

Next we'll need to select the server to install ADDS to. This lab setup currently has only one server named ADC and that's the one we'll use for this installation. Setting recognizable host names helps for tasks such as this. Important to note: to use this server as a domain controller it will need a static IP address. This IP address is an APIPA address and is changed for a static /24 IP later.

- ☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address
ADC	169.254.189.36

Roles

- ☐ Active Directory Certificate Services
- ☒ Active Directory Domain Services
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☒ DHCP Server
- ☒ DNS Server
- ☐ Fax Server
- ☒ File and Storage Services (1 of 12 installed)
- ☐ Host Guardian Service
- ☐ Hyper-V

The roles that we'll be installing are pictured here. Select Active Directory Domain Services and DNS server to facilitate Active Directory. Name resolution will be necessary for the domain. This server will also act as the DHCP server, which will be configured and detailed later.

After selecting each role a dialog box will appear detailing additional services or features that are needed for that role. By default a checkbox is selected to "Include management tools (if applicable)". This default is kept for each role installed here.

Features

- ☐ .NET Framework 3.5 Features
- ☒ .NET Framework 4.8 Features (2 of 7 installed)
- ☐ Background Intelligent Transfer Service (BITS)
- ☐ BitLocker Drive Encryption
- ☐ BitLocker Network Unlock
- ☐ BranchCache
- ☐ Client for NFS
- ☐ Containers
- ☐ Data Center Bridging
- ☐ Direct Play
- ☐ Enhanced Storage
- ☐ Failover Clustering
- ☒ Group Policy Management

Under Features Group Policy Management should be selected by default. Just to be safe, double check that this feature is selected before moving on to the Confirmation screen.

On the Confirmation screen the roles and features can be verified before installation. We have further configuration to do, so we won't select the "Restart the destination server option here". Now we can click install and wait for it to complete.

Confirm installation selections

Before You Begin

Installation Type

Server Selection

Server Roles

Features

AD DS

DHCP Server

DNS Server

Confirmation

Results

To install the following roles, role services, or features on selected server, click Install.

☐ Restart the destination server automatically if required

Optional features (such as administration tools) might be displayed on this page because they selected automatically. If you do not want to install these optional features, click Previous to close boxes.

Active Directory Domain Services

DHCP Server

DNS Server

Group Policy Management

Remote Server Administration Tools

 Role Administration Tools

 AD DS and AD LDS Tools

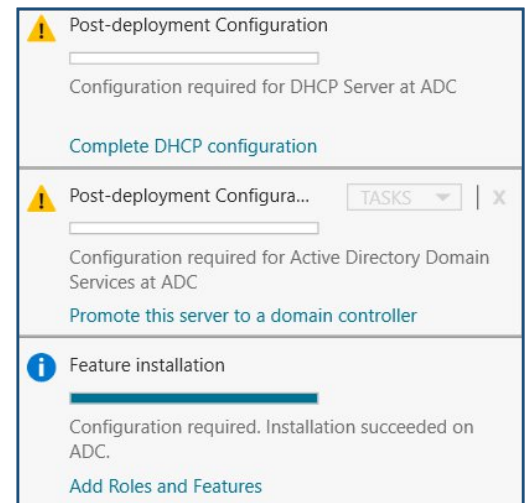
 Active Directory module for Windows PowerShell

 AD DS Tools

 Active Directory Administrative Center

 AD DS Snap-Ins and Command-Line Tools

After finishing the install we head back to the Server Dashboard. Next to Manage at the top is a notification showing that we need to configure our DHCP and ADDS services. Right now we need to promote the server to Domain Controller to continue the ADDS configuration. We can click the blue text in the notification to proceed to that step and start the wizard.



The first step will be to add a new forest since there is not a domain to add the server to. This domain is going to be called "laborg.local". On the following screen we need to set a password for DSRM, but otherwise the default options will be kept. Going forward the rest of the wizard will use the default settings as well, but will be elaborated on where relevant.

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

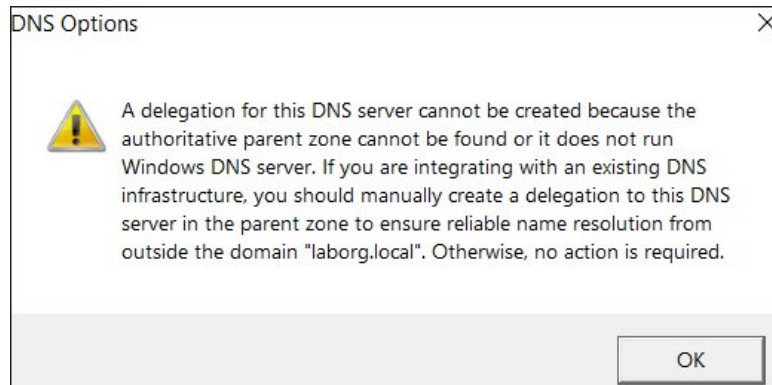
☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

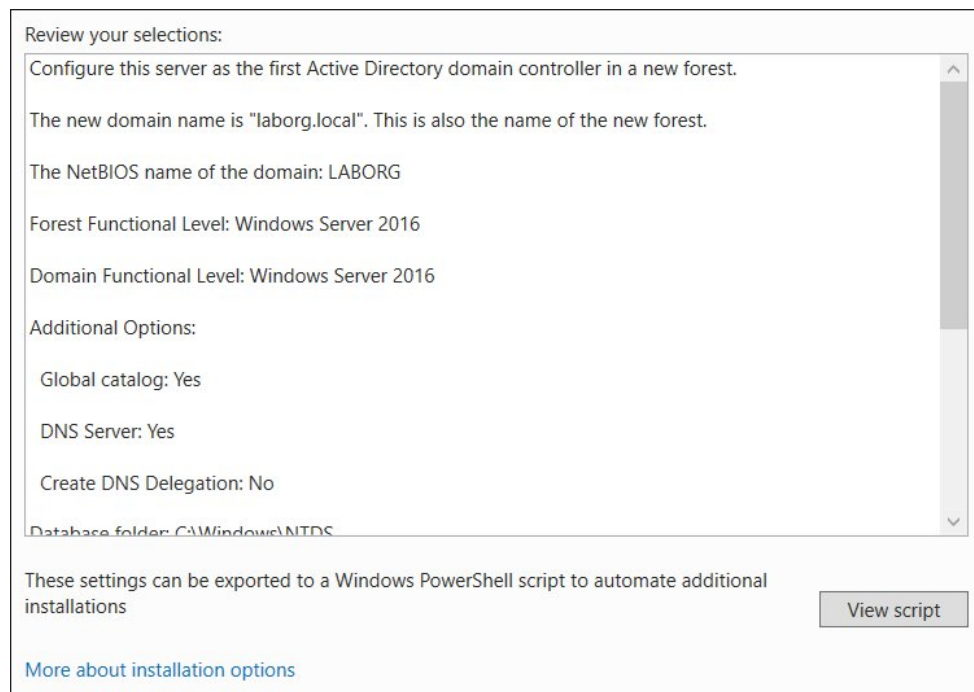
Password: ••••••••

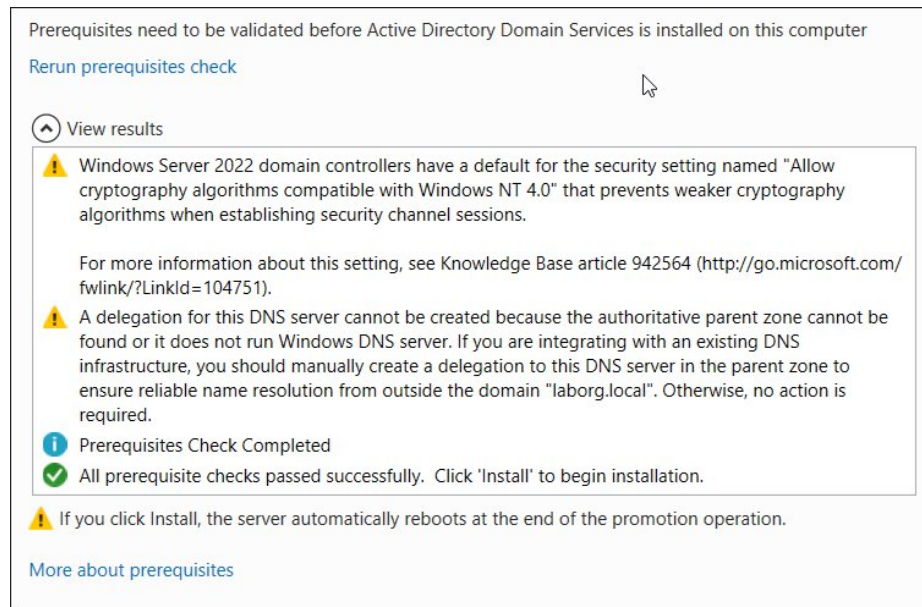
Confirm password: ••••••••

At the top of the following screen will be a warning concerning DNS. Clicking the link will open the dialog box pictured below. This server isn't being added to a network that already has DNS implemented. By default the checkbox to create a DNS delegation will not be selected and since we don't need to do such, we can proceed to the next step.



The next steps will be to verify the NetBIOS name and paths relevant to Active Directory. Both of these will be left as the default. However, for the NetBIOS name it is important to double check that it matches the domain name. In this example the NetBIOS name should be "LABORG" for the domain "laborg.local". After accepting the defaults for the NetBIOS name and paths, we can review our selections before proceeding to the installation.





Next the wizard will perform a check to make sure all the prerequisites for the Domain Controller have been met. In the above screenshot the check has passed which can be seen next to the green check mark towards at the bottom of the screen. Even though the checks have passed, there are some important notifications to be aware of:

- 1) Windows Server 2022 defaults to not allow “weaker cryptography algorithms when establishing security channel sessions. This setting can be changed for to facilitate older algorithms and legacy systems, but is not recommended. An excerpt from Microsoft’s documentation explains this further:

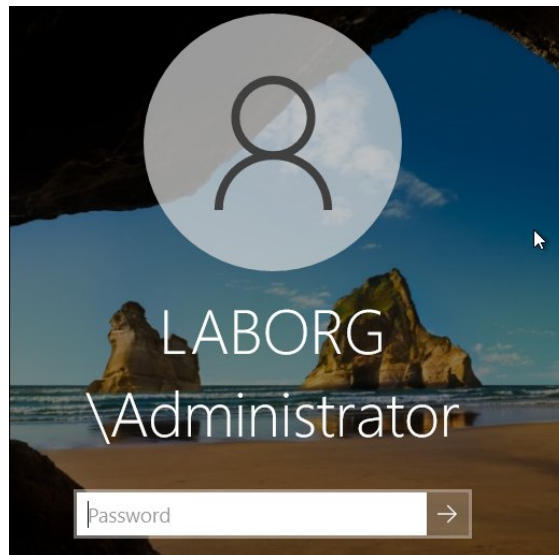
Context & Best Practices

By default, Windows Server 2008 or later prohibits clients running non-Microsoft operating systems or Windows NT 4.0 operating systems to establish secure channels using weak Windows NT 4.0–style cryptography algorithms. Any security-channel-dependent operation that is initiated by clients running older versions of the Windows operating system or running non-Microsoft operating systems that do not support strong cryptographic algorithms will fail against a domain controller that runs Windows Server 2008, Windows Server 2008 R2 or Windows Server 2012 with default settings.

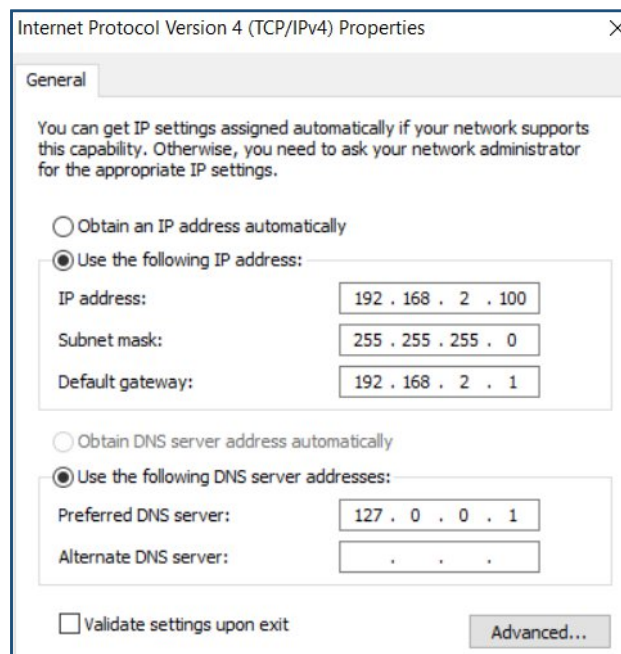
Windows Server 2008 R2 and later do not support trust relationships with Windows NT 4.0 even when using the NT4Crypto setting. This limitation includes but is not limited to the following secure channel operations: - Establishing and maintaining trust relationships - Domain Join - Domain authentication - SMB sessions

- 2) This warning is the same warning about DNS delegation from earlier. As explained earlier this can be disregarded for this lab.

After the installation process the server will reboot and the changes can be confirmed with the domain name appearing on the login screen as shown in the following screenshot.



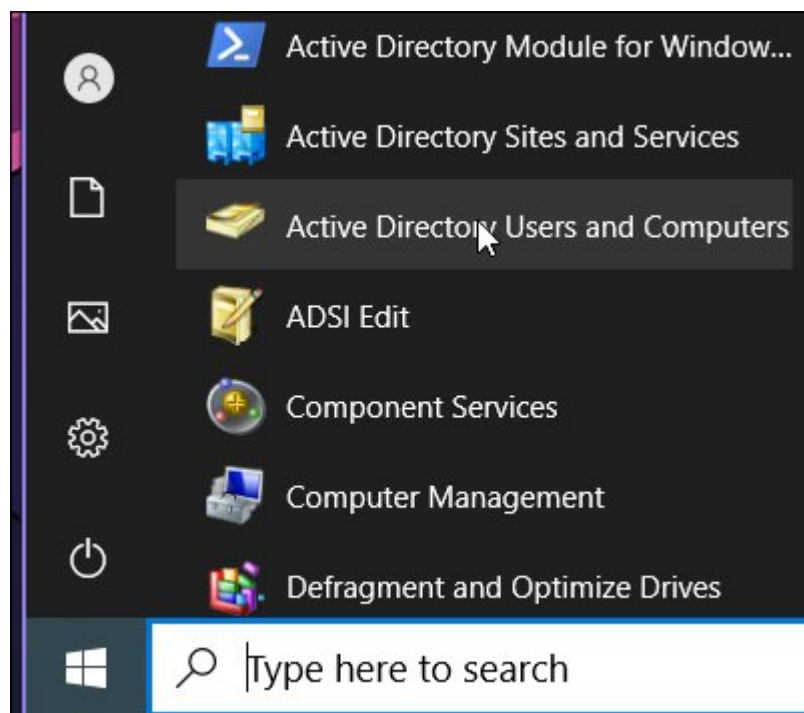
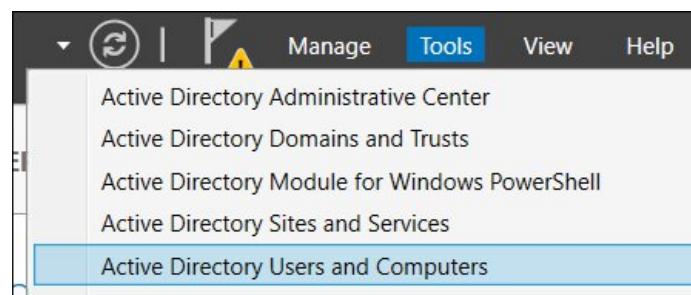
Earlier in the lab set up it was mentioned that the server currently held an APIPA address. This type of IP address is assigned automatically by the system to allow it to communicate and receive information from other locally connected devices. This IP address will be changed so that it can be properly routed later and will be necessary when setting up DHCP later. The network address will be 192.168.2.0/24. The IP address configuration is pictured here:



Create New Admin User

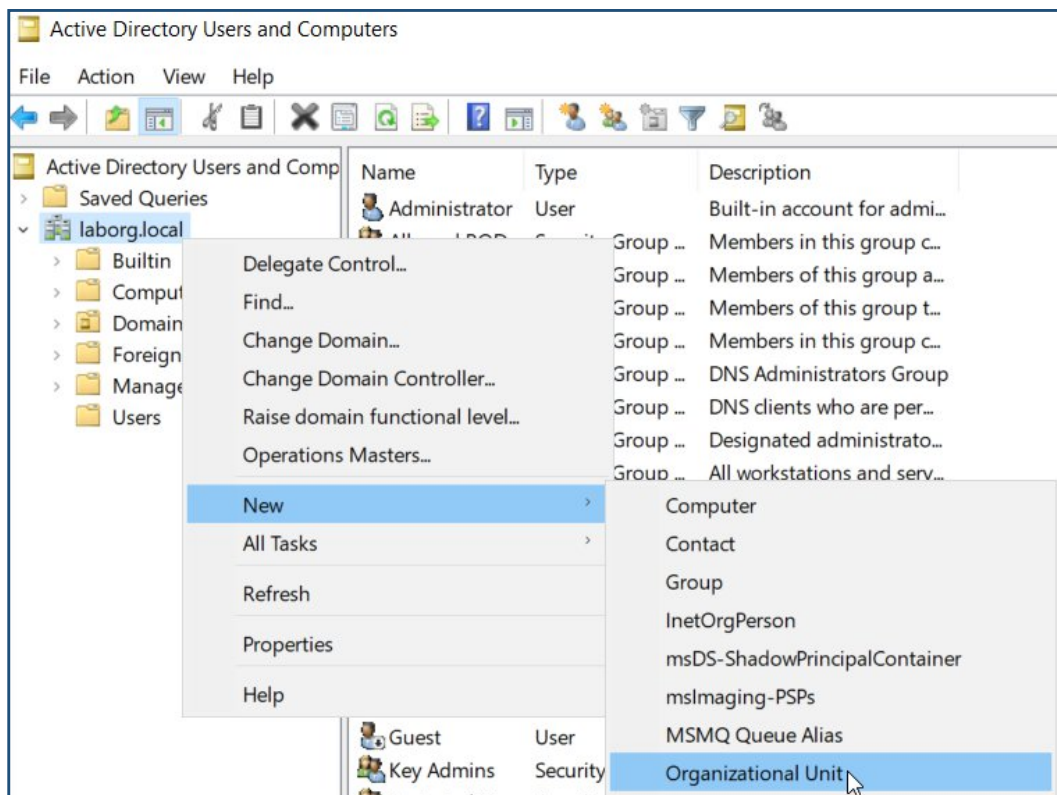
Next we'll create a new user and add them to the Administrators security group. This user account will then be used to join a workstation to the domain. This is also a security best practice, as using the default Administrator account can pose a significant risk. Some attacks might target the identifier for the default Administrator account for example. By creating a new user that can be mitigated and the user's permissions can be further refined if necessary. This can be taken a step further by creating a new security group for the domain's administrators, allowing for more granular control through Group Policy.

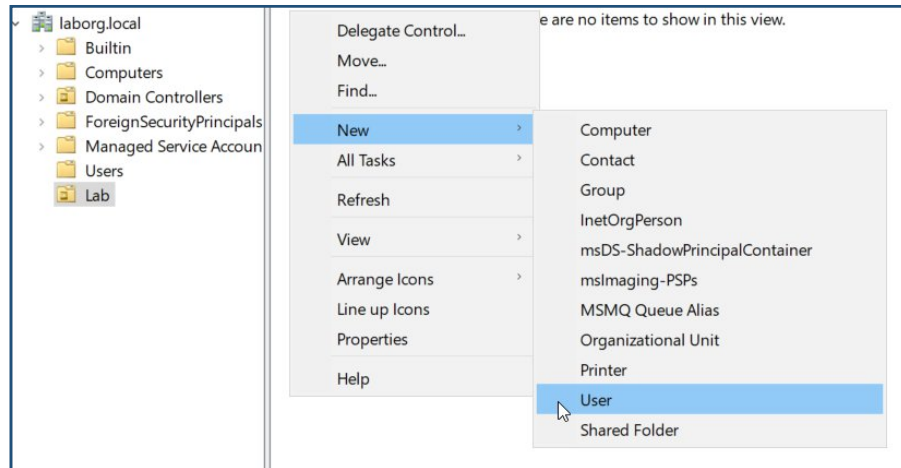
The new user will be created using Active Directory Users and Computers. This can be accessed from the Server Management dashboard tool bar or through the start menu as shown here:



The left pane shows the organization of the domain, each directory acting as an Organizational Unit (OU) for the domain tree. These OUs are useful for grouping users and computers within the domain and can be used for assigning policies (or GPOs – Group Policy Objects). Setting up OUs and Group Policy can get very complex but will be kept simple for the lab setup process. The right pane shows what resources are within the selected OU or domain.

Before creating our user we'll create a new OU named "Lab" for our domain's users. Right-clicking the domain name in the left pane will let us do this as shown in the screenshot below. We can also create new users, computers, or security groups using the action bar or toolbar.

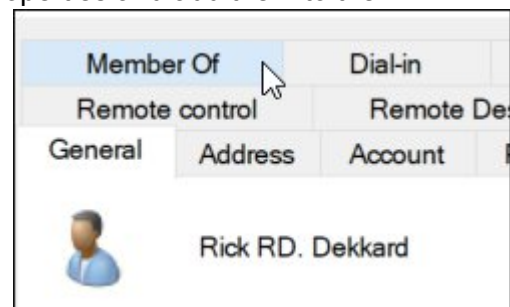




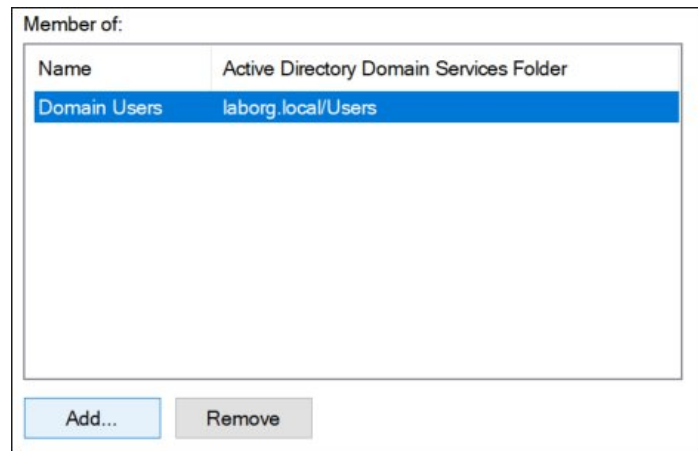
To create the user we can right-click the Lab OU in the left pane, navigate to New and then select User from the menu. This user is named Rick Dekkard with the username **dekkard**. Typing the username in the first box should populate the second box labeled “User logon name (pre-Windows 2000)” but if not the boxes should match before proceeding. We’ll also need to set a password for the user. For now we’ll uncheck the “User must change password at next logon” which is set by default. This function will be demonstrated after joining the workstation to the domain.

 A screenshot of the 'New User' wizard in Windows. The 'Full name' field contains 'Rick RD. Dekkard'. The 'User logon name' field contains 'dekkard' and the domain dropdown is '@laborg.local'. The 'User logon name (pre-Windows 2000)' section has 'LABORG\' in the first box and 'dekkard' in the second box. On the right, the 'Password' and 'Confirm' fields are masked with dots. Below these, the 'User must change password at next logon' checkbox is unchecked and highlighted by a mouse cursor. Other unchecked options include 'User cannot change password', 'Password never expires', and 'Account is disabled'. At the bottom are '< Back', 'Next >', and 'Cancel' buttons.

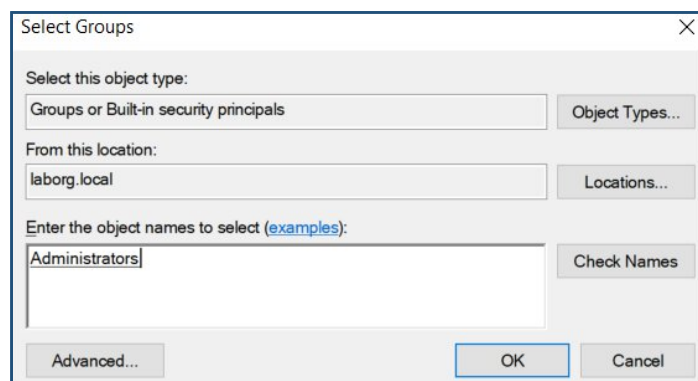
Once the user is created we can go to the user properties and add them to the Administrator security group. To get to the user properties the process is the same as opening the properties of a file or directory in Windows Explorer. Simply right-click the user and select properties from the menu. There are many tabs at the top of the properties window. The Member Of tab will allow us to edit the security groups the user belongs to.



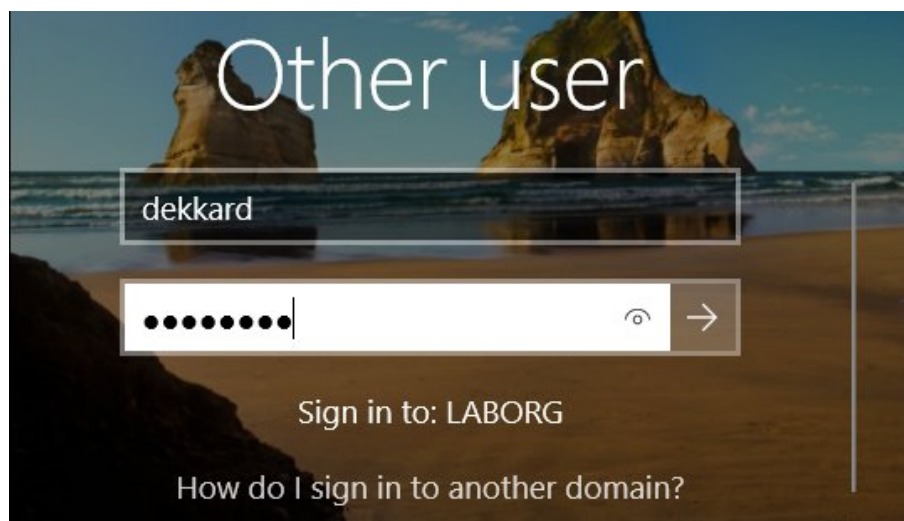
In the right screenshot we can see the list of groups that Dekkard belongs to. By default it only includes the Domain Users group. By clicking the Add button we can move to the next step.



This is the Select Groups screen. The text box on the bottom allows us to search for the group we need to add. Typing "Admin" and then clicking Check Names beside the box will auto-complete the group name if it's found. If there are multiple groups with similar names you made need to type the name exactly instead of searching.

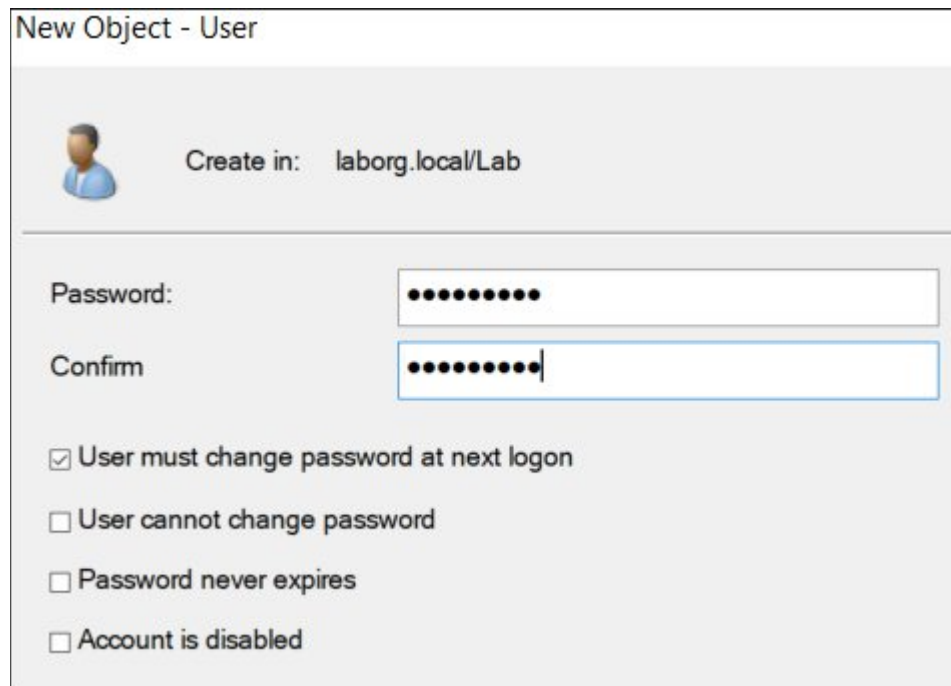


Clicking OK will show the updated groups but the changes won't be applied until clicking OK or Apply at the bottom of the window. Once this is finished we can logout and log in as the new user using the Other User option on the login screen.



Creating Standard User

Now we'll create a regular user and will leave the default security group (Domain Users). In a real world scenario, this user would most like be added to a group depending on their department or responsibilities within the organization (i.e. Marketing, Accounting, etc.).



New Object - User

Create in: laborg.local/Lab

Password:

Confirm:

☒ User must change password at next logon

☐ User cannot change password

☐ Password never expires

☐ Account is disabled

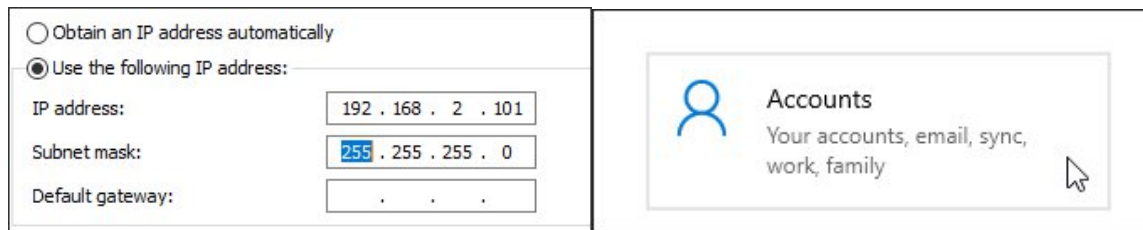
This user's name is Stan Marsh (smarsh@laborg.local). We'll also leave the option to force a password change at next login selected. After the workstation is joined to the domain we'll use this user account to log in.

Name	Type	Description
 Rick RD. Dek...	User	
 Stan Marsh	User	

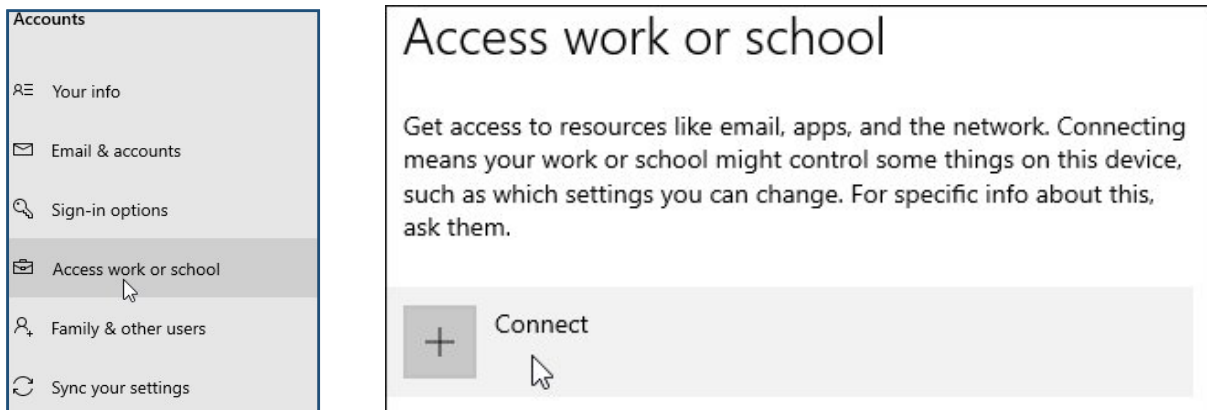
Joining ADDS Domain

Operating System: Windows 10 Pro

This process can be replicated using Windows 11 Pro, but may look slightly different. Joining the domain can be done during or post-installation. In this scenario we'll be joining the domain post-installation. In order to do this the machine will need a static IP address within the domain (192.168.2.101/24 in this case). This will allow the machine to communicate with the server. Then we open Settings and can get to the Accounts section by clicking the Accounts button as seen below.



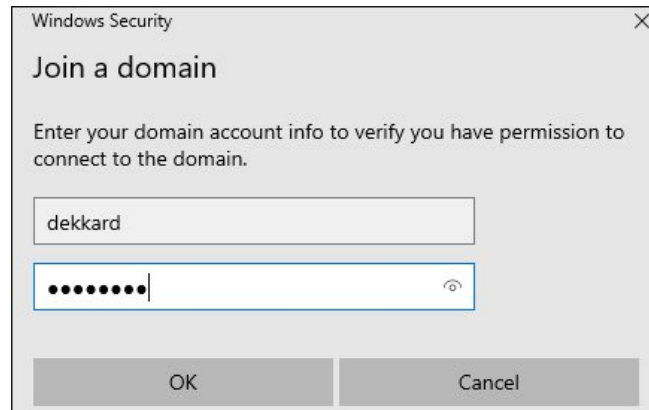
In the left pane of the window are the different account settings. To connect to the domain we'll navigate to the section labeled "Access work or school". On this page we then click connect to start the process.



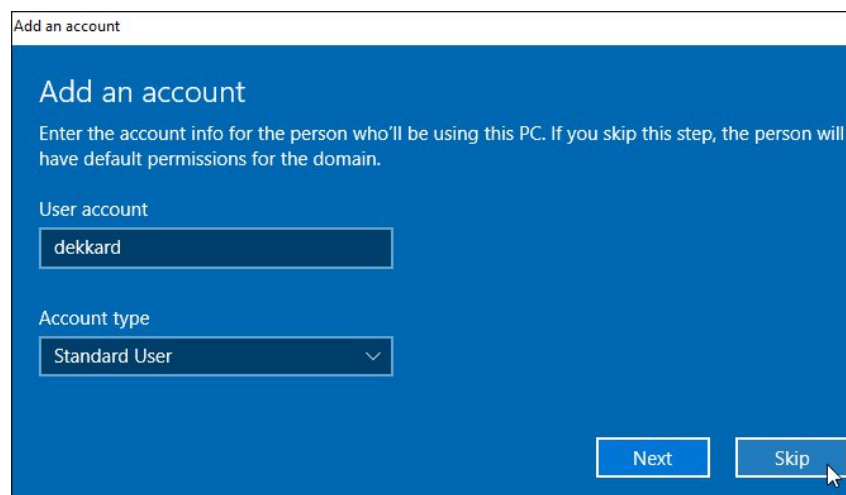
A window will appear with a text box where you can enter an email address. Instead, click the blue text at the bottom of the window that says "Join this device to a local Active Directory domain."



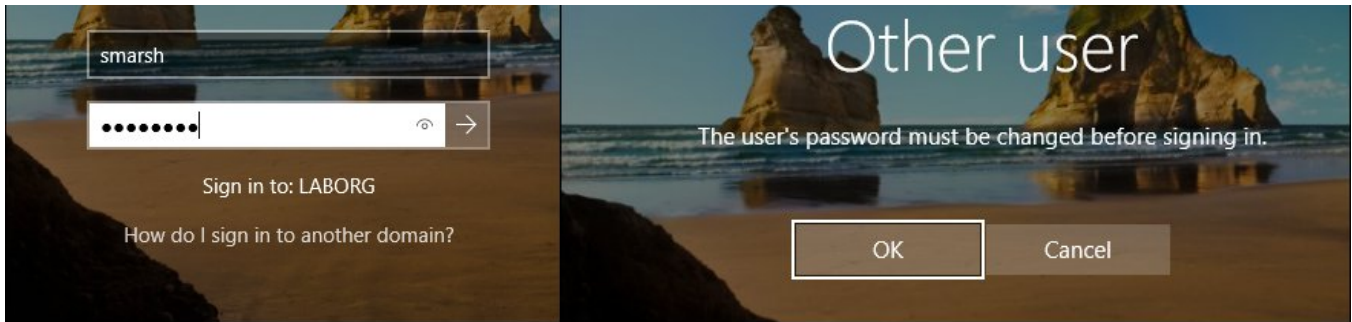
A blue window will appear (pictured below) with a text box where you can enter the domain name. After entering **laborg.local** and clicking next we'll be prompted for a username and password. As it says, the account must have permissions to connect to the domain. We'll use Dekkard's credentials since he has admin privileges.



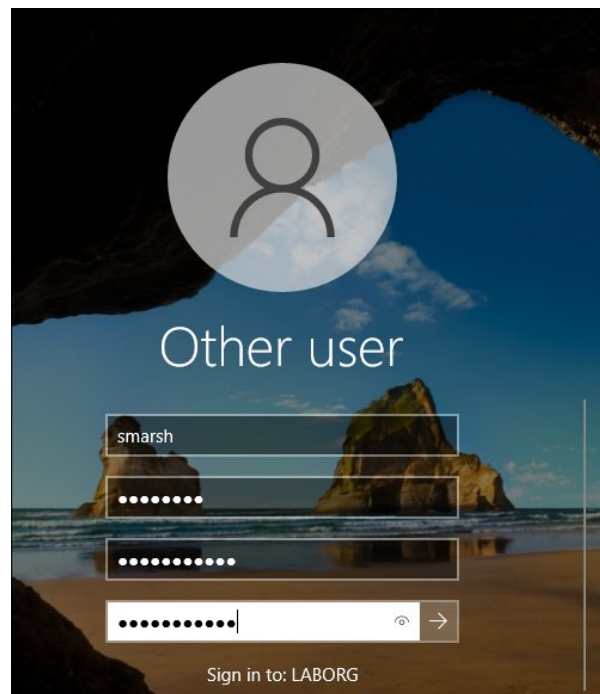
After successful authentication you can add the account to the workstation and set the account type. We can skip this for now since we'll be logging in as the standard user Stan Marsh after the device reboots. It's worth noting that skipping also means the permissions for the user will default to the domain defined permissions (i.e. Group Policy). Once skipping we'll be prompted to reboot the system and can log into the workstation.



Change User Password At Next Login



Now that the workstation has been rebooted we can log in using a domain user. Selecting “Other user” at the login screen and entering the credentials for **smarsh** will give us the prompt show above. When we created the user we used the option to force a password change at the next login. This is useful in situations where a new user is provisioned (new hires for example). The user can then login with some default credential and set their own *private* password unknown to anyone else.



Conclusion

Having completed everything will leave you with a domain containing one server and one workstation. The workstation can be logged into using any user that is created in Active Directory (unless otherwise specified via policy). Some recommended tasks to experiment with are:

- Password Policies
- Security Groups/Permissions
- Using Powershell to automate user creation

I have further plans to add DHCP, add an Ubuntu machine for monitoring the network, and setup pfSense to act as a firewall and router for the network. This lab is already somewhat lengthy so those will be detailed separately at a later date.

pfSense



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